MATH-1014 T10B: SOLUTIONS

Advanced Convergence Tests and Strategies

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Part A: Alternating Series & Absolute vs. Conditional Convergence

1. **Solution:** For absolute convergence: $\sum \frac{n}{n^2+4}$ diverges by Limit Comparison Test with $\sum \frac{1}{n}$. For conditional convergence: $b_n = \frac{n}{n^2+4}$ is decreasing and $\lim_{n\to\infty} b_n = 0$. By the Alternating Series Test (AST), it converges. **Answer:** Conditionally convergent.

2. **Solution:** Rewrite the term: $(-1)^n(\sqrt{n+1} - \sqrt{n}) = \frac{(-1)^n}{\sqrt{n+1}+\sqrt{n}}$. The absolute series $\sum \frac{1}{\sqrt{n+1}+\sqrt{n}}$ diverges (LCT with $1/\sqrt{n}$). Since $b_n = \frac{1}{\sqrt{n+1}+\sqrt{n}}$ decreases to 0, it converges by AST. **Answer:** Conditionally convergent.

3. **Solution:** Let $a_n = e - (1 + \frac{1}{n})^n$. As $n \to \infty$, $a_n \to 0$. Using Taylor expansion, $a_n \approx \frac{e}{2n}$. Thus, $\sum |a_n|$ diverges by comparison to $\sum \frac{1}{n}$. Since a_n decreases to 0, AST applies. **Answer:** Conditionally convergent.

4. **Solution:** Multiply numerator and denominator by $\sqrt{n}-(-1)^n$: $\frac{(-1)^n}{\sqrt{n}+(-1)^n} = \frac{(-1)^n\sqrt{n}-1}{n-1} = \frac{(-1)^n\sqrt{n}}{n-1} - \frac{1}{n-1}$. The first part converges by AST, but the second part $\sum \frac{-1}{n-1}$ is a divergent harmonic series. **Answer:** Diverges.

5. **Solution:** $\sum |e^{-2n} \sin n| \leq \sum e^{-2n} = \sum (\frac{1}{e^2})^n$. Since $1/e^2 < 1$, the bounding geometric series converges. By the Direct Comparison Test, the series converges absolutely. **Answer:** Absolutely convergent.

6. **Solution:** The absolute series $\sum \frac{1}{\sqrt{n+1}}$ is a p-series with $p = 1/2 \leq 1$, which diverges. However, $b_n = \frac{1}{\sqrt{n+1}}$ is positive, decreasing, and $\lim b_n = 0$, so it converges by AST. **Answer:** Conditionally convergent.

Part B: The Ratio and Root Tests

7. **Solution:** Use Ratio Test: $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)^3}{3^{n+1}} \cdot \frac{3^n}{n^3} = \frac{1}{3} \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^3 = \frac{1}{3} < 1$. **Answer:** Absolutely convergent.

8. **Solution:** Use Root Test: $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{2n+3}{3n+2} = \frac{2}{3} < 1$. **Answer:** Converges.

9. **Solution:** Use Root Test: $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^n = \lim_{n\to\infty} \frac{1}{(1+1/n)^n} = \frac{1}{e} < 1$. **Answer:** Converges.

10. **Solution:** Use Root Test: $\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} < 1$. **Answer:** Converges.

11. **Solution:** Simplify to $\sum \frac{10^n n!}{n^n}$. Use Ratio Test: $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{10^{n+1}(n+1)!}{(n+1)^{n+1}} \frac{n^n}{10^n n!} = \lim_{n\to\infty} 10 \left(\frac{n}{n+1}\right)^n = \frac{10}{e} \approx 3.68 > 1$. **Answer:** Diverges.

Part C: Comparison Tests & Asymptotic Analysis

12. **Solution:** For large n, $a_n \approx \frac{\sqrt{n^3}}{3n^3} = \frac{1}{3n^{3/2}}$. By Limit Comparison Test with $b_n = \frac{1}{n^{3/2}}$ (a convergent p-series with $p = 1.5 > 1$), the series converges. **Answer:** Converges.

13. **Solution:** Note that for $n > e^{e^2}$, $\ln n > e^2$, so $(\ln n)^{\ln n} > (e^2)^{\ln n} = n^2$. Thus, $a_n < \frac{1}{n^2}$. By Direct Comparison Test, it converges. **Answer:** Converges.

14. **Solution:** Using Maclaurin expansion, $\sin(1/n) \approx 1/n - \frac{1}{6n^3}$. Thus, $a_n = \frac{1}{n} - \sin(1/n) \approx \frac{1}{6n^3}$. By LCT with $b_n = 1/n^3$, it converges. **Answer:** Converges.

15. **Solution:** For large n, $a_n \approx \frac{\sqrt{n}}{n^2} = \frac{1}{n^{3/2}}$. By LCT with convergent p-series ($p = 1.5 > 1$), it converges. **Answer:** Converges.

16. **Solution:** For $n \geq 3$, $\ln n > 1$, so $\frac{\ln n}{2n+3} > \frac{1}{2n+3}$. Since $\sum \frac{1}{2n+3}$ diverges, the series diverges by Comparison Test. **Answer:** Diverges.

17. **Solution:** (i) Absolutely converges ($p = 2.5$). (ii) Converges (Ratio test $\to 0$). (iii) Converges (Absolute value bounded by $1/n^2$). (iv) Converges (Root test $\to 1/4$). **Answer:** (i), (ii), (iii), and (iv) are all convergent.

Part D: The Integral Test & Improper Integrals

18. **Solution:** Let $f(x) = \frac{1}{x \ln x \sqrt{\ln(\ln x)}}$. Integral test: let $u = \ln(\ln x)$, $du = \frac{dx}{x \ln x}$. $\int \frac{1}{\sqrt{u}} du = 2\sqrt{u} \to \infty$. **Answer:** Diverges.

19. **Solution:** Integral test: let $u = \ln x$, $du = dx/x$. Integral becomes $\int u^{-p} du$, which converges if and only if $p > 1$. **Answer:** $p > 1$.

20. **Solution:** Near $x = 0$, integrand $\approx x^{-1/2}$ (converges). Near $x = \infty$, bounded by e^{-x} (converges). $\int_0^\infty x^{-1/2} e^{-x} dx = \Gamma(1/2) = \sqrt{\pi}$. **Answer:** Converges (to $\sqrt{\pi}$).

21. **Solution:** Integration by parts ($u = \ln x, dv = x^{-2} dx$): $\int_1^\infty \frac{\ln x}{x^2} dx = \left[-\frac{\ln x}{x}\right]_1^\infty + \int_1^\infty x^{-2} dx = 0 + \left[-\frac{1}{x}\right]_1^\infty = 1$. **Answer:** Converges exactly to 1.

22. **Solution:** Trig substitution $x = 2\sec\theta$. $dx = 2\sec\theta\tan\theta d\theta$. Integral becomes $\int_0^{\pi/3} \frac{3(2\sec\theta\tan\theta)}{2\sec\theta(2\tan\theta)} d\theta = \frac{3}{2}\int_0^{\pi/3} 1 d\theta = \frac{\pi}{2}$. **Answer:** $\pi/2$.

23. **Solution:** Let $u = 1/x, du = -1/x^2 dx$. Bounds $x \in [2, \infty) \to u \in [1/2, 0]$. $\int_{1/2}^0 \frac{-5}{\sqrt{1/u^2+5}} du = \int_0^{1/2} \frac{5u}{\sqrt{1+5u^2}} du = \frac{1}{2}[\sqrt{1+5u^2}]_0^{1/2} = \frac{1}{2}(\sqrt{9/4} - 1) = 1/4$. **Answer:** $1/4$.

24. **Solution:** (a) Sub $x = \tan\theta$: $\int_0^{\pi/2} \cos^2\theta d\theta = \pi/4$. (b) Integral is $[-4\sqrt{4-x^2}]_0^2 = 0 - (-8) = 8$. **Answer:** (a) $\pi/4$; (b) 8.

25. **Solution:** Integral evaluates to $[-\frac{5}{2}\ln(\cos x)]_0^{\pi/2}$. As $x \to \pi/2$, $\cos x \to 0$, so $\ln(\cos x) \to -\infty$. **Answer:** Diverges.

26. **Solution:** Combine: $\int_0^R \frac{2x^2+6x-Cx^2-2C}{(x^2+2)(x+3)} dx$. For limit at ∞ to converge, the x^2 coefficient must vanish to eliminate logarithmic growth. $2 - C = 0 \implies C = 2$. **Answer:** $C = 2$.

27. **Solution:** As $x \to \infty$, $\frac{3x^2}{2x^3} - \frac{kx}{2x^2} = \frac{3}{2x} - \frac{k}{2x}$. To avoid divergence, $\frac{3-k}{2} = 0$. **Answer:** $k = 3$.

28. **Solution:** (a) Converges (bounded by e/x^2). (b) Converges (behaves as $1/x^3$). (c) Converges (exponential decay). (d) Diverges, because $\frac{1}{(\ln x)^2} > \frac{1}{x}$ for sufficiently large x, and $\int \frac{1}{x}$ diverges. **Answer:** (d) is divergent.

29. **Solution:** (i) Diverges ($1/\sqrt{x}$). (ii) Converges ($2\ln(x)/x^2$). (iii) Diverges ($\frac{2^x}{x+2^x} \to 1 \neq 0$). (iv) Diverges ($1/(x\ln x)$). (v) Diverges ($2/(x\ln x)$). **Answer:** (ii) is convergent.

Part E: Exact Sums & Limits

30. **Solution:** Riemann Sum for $\int_0^1 \frac{x}{1+x^2} dx$. Let $u = 1 + x^2$, $du = 2x dx$. Integral is $\frac{1}{2}[\ln(1+x^2)]_0^1 = \frac{1}{2}\ln 2$. **Answer:** $\frac{\ln 2}{2}$.

31. **Solution:** Telescoping series. $\ln(1 - 1/n^2) = \ln(\frac{n-1}{n}) + \ln(\frac{n+1}{n})$. Adding terms from $n = 2$ to N causes massive cancellation, leaving $\ln(1/2) + \ln((N+1)/N) \to -\ln 2$. **Answer:** $-\ln 2$.

32. Solution: (a) As $n \to \infty$, angle $\to \pi/2$. $\cos(\pi/2) = 0$. (b) $\frac{\ln n}{\ln n + \ln 3} = \frac{1}{1+\ln 3/\ln n} \to 1$. **Answer:** (a) 0; (b) 1.

33. Solution: Set $L = 6 - 4/L \implies L^2 - 6L + 4 = 0 \implies L = \frac{6\pm\sqrt{20}}{2} = 3 \pm \sqrt{5}$. Since $a_1 = 2$ and $a_2 = 4$, the sequence is increasing. It converges to the larger root. **Answer:** $3 + \sqrt{5}$.

34. Solution: (i) $\to 1$. (ii) $\to \infty$. (iii) Rationalize: $\to 1$. (iv) $\to 0$. (v) $\to \infty$. **Answer:** (i), (iii), (iv) converge to finite limits.

35. Solution: (a) Split into $\sum 12(1/2)^n + \sum 4(1/3)^n = 12(1) + 4(1/2) = 14$.
(b) $\sum_{n=1}^{\infty} n(1/2)^{n-1} = \frac{1}{(1-1/2)^2} = 4$. **Answer:** (a) 14; (b) 4.

36. Solution: Let $S_N = \sum_{n=1}^{N}((1/2)^{1/(n+1)} - (1/2)^{1/n})$. This telescopes to $(1/2)^{1/(N+1)} - (1/2)^1$. As $N \to \infty$, $(1/2)^0 - 1/2 = 1 - 1/2 = 1/2$. **Answer:** $1/2$.

37. Solution: AST error bound: $|R_n| \leq b_{n+1} = \frac{1}{(n+1)^3}$. Let ϵ be the specified accuracy. We require $\frac{1}{(n+1)^3} \leq \epsilon \implies n \geq \lceil \epsilon^{-1/3} \rceil - 1$. **Answer:** $n \geq \lceil \epsilon^{-1/3} \rceil - 1$.

Part F: Applications of Improper Integrals

38. Solution: $V = 2\pi \int_0^\infty x^3 e^{-x^2} dx$. Let $u = x^2, du = 2xdx$. $V = \pi \int_0^\infty ue^{-u} du = \pi(1) = \pi$. **Answer:** π.

39. Solution: The function $\pi x^4 e^{-2x^2}$ decays exponentially, which is faster than any polynomial $1/x^p$. Thus the integral evaluates to a finite constant. (Specifically, it is $\frac{3\pi}{16}\sqrt{\pi/2}$). **Answer:** Yes, it is finite.

40. Solution: $\int_0^\infty \frac{4x}{(x^2+1)^3} dx = \left[-\frac{1}{(x^2+1)^2}\right]_0^\infty = 0 - (-1) = 1$. **Answer:** 1.

41. Solution: Cylindrical shells: $V = \int_0^\infty 2\pi x(9e^{-3x})dx = 18\pi \int_0^\infty xe^{-3x} dx = 18\pi \left(\frac{1}{9}\right) = 2\pi$. **Answer:** 2π.

42. Solution: Disks: $V = \pi \int_0^\infty (2xe^{-x^3/6})^2 dx = \pi \int_0^\infty 4x^2 e^{-x^3/3} dx$. Let $u = x^3/3, du = x^2 dx$. $V = 4\pi \int_0^\infty e^{-u} du = 4\pi$. **Answer:** 4π.

43. Solution: $V = \int_0^\infty \frac{64\pi}{e^{2x}+4} dx$. Let $u = e^x, du = e^x dx \implies dx = du/u$. Integral becomes $64\pi \int_1^\infty \frac{1}{u(u^2+4)} du$. Using partial fractions $(\frac{1}{4u} - \frac{u}{4(u^2+4)})$, this evaluates to $8\pi \ln(5/2)$. The second integral clearly converges as its integrand is $\approx 16\pi xe^{-x}$. **Answer:** Exact value is $8\pi \ln(5/2)$. Yes, the second integral converges.

Part G: Power Series & Taylor/Maclaurin Series

44. Solution: (a) Ratio test yields $\frac{|x-2|^2}{16} < 1 \implies |x-2| < 4 \implies (-2, 6)$.
(b) At $x = -2$ and $x = 6$, series becomes $\sum \frac{(-1)^n}{2n+1}$, which converges conditionally (AST).
(c) Integrate term-by-term: $\int_2^3 (x-2)^{2n} dx = \frac{1^{2n+1}}{2n+1}$. The exact value is $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2 16^n}$.
Answer: (a) $(-2, 6)$; (b) Both conditionally converge; (c) $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2 16^n}$.

45. Solution: (a) Ratio test gives $\frac{|x-2|}{4} < 1 \implies (-2, 6)$.
(b) At $x = 6$, $\sum \frac{(-1)^n}{n+2}$ converges (AST). At $x = -2$, $\sum \frac{1}{n+2}$ diverges (Harmonic).
(c) $H(x) = \sum \frac{(-1)^n}{(n+2)4^n}(x-2)^{n+2} \implies H'(x) = \sum \frac{(-1)^n}{4^n}(x-2)^{n+1}$. Evaluating at $x = 4$ gives $\sum_{n=0}^{\infty} 2(-1/2)^n = 2(\frac{1}{1-(-1/2)}) = \frac{4}{3}$. **Answer:** (a) $(-2, 6)$; (b) Converges at $x = 6$, diverges at $x = -2$; (c) $4/3$.

46. Solution: Simplify $f(x) = 2\ln(2 + x^2) = 2\ln 2 + 2\ln(1 + x^2/2)$. The Maclaurin series for $\ln(1+u)$ is $\sum(-1)^{n-1}u^n/n$. Substituting $u = x^2/2$, all resulting powers of x will be even (e.g., x^2, x^4, x^6). Therefore, the coefficient of x^5 is 0. **Answer:** 0.

47. Solution: Use Binomial Series $(1+u)^k = 1 + ku + \frac{k(k-1)}{2!}u^2 + \ldots$ where $u = 8x^2$ and $k = 3/2$. The x^4 term comes from the u^2 term: $\frac{(3/2)(1/2)}{2}(8x^2)^2 = \frac{3}{8}(64x^4) = 24x^4$. **Answer:** 24.